

An Internship Report
Of
**Artificial Intelligence Internship
Programme**

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech (Computer Science and Engineering)



By

Roll Number of Student: 2410030383

Name of Student: SHAGUN JAIN

Batch: 2024-2028

CANDIDATE'S DECLARATION

I, Shagun Jain, do hereby declare that the internship report titled “Artificial Intelligence Internship Programme” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: Shagun Jain

Roll No.: 2410030383

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature:-

Student Name: Shagun Jain

Roll No.: 2410030383

TABLE OF CONTENTS

1. Candidate's Declaration.....	i
2. Acknowledgement.....	ii
3. Internship Completion Certificate.....	-
4. Project Description.....	1
4.1 Introduction.....	1
4.2 Organisation Profile	1
4.3 Problem Statement	2
4.4 Project Objectives	2
4.5 Scope of the Project.....	2
4.6 Technologies and Tools Used	2
4.7 System Architecture	3
4.8 Methodology	3
4.9 Expected Outcomes.....	4
4.10 Certificates of Completion and Communication Proof	4
5. Bibliography/References	6

3. INTERNSHIP COMPLETION CERTIFICATE



Internship Offer Letter



INTERNSHIP OFFER LETTER

Dear Shagun Jain,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Artificial Intelligence Intern
- Location & Mode: Pan India, Hybrid
- Duration : 22/07/2026 to 22/08/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.

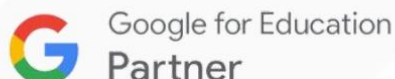
Best regards,

Sincerely,

Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 22/07/2026




Dr. Anurag Shrivastava
Talent Acquisition Manager



4. PROJECT DESCRIPTION

4.1 Introduction

Artificial Intelligence (AI) has rapidly become one of the most transformative technologies across industries, enabling systems to learn from data, recognise patterns, and make intelligent decisions with minimal human intervention. To gain practical, industry-oriented exposure to this domain, I undertook a one-month Artificial Intelligence Internship with Codec Technologies Pvt. Ltd., conducted under the National Internship Portal of the Ministry of Education (AICTE), from 22/07/2026 to 22/08/2026.

The internship followed a structured four-week curriculum covering the foundations of AI and Python for data handling, core Machine Learning concepts, Neural Networks and Deep Learning, and advanced applications such as Natural Language Processing (NLP) and Reinforcement Learning, culminating in a capstone project. For my capstone project, I chose to build a Sentiment Analysis Model, an NLP-based application that automatically classifies text as expressing positive, negative, or neutral sentiment. This report documents the internship experience, the organisation, and the project undertaken in detail.

4.2 Organization Profile

Codec Technologies Pvt. Ltd. is a global platform dedicated to empowering learners and connecting diverse talent with meaningful career opportunities worldwide. The organisation delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth for its clients and partners.

Registered internship provider on the National Internship Portal (AICTE ID: CORPORATE6759d549ce59e1733940553), Ministry of Education, Government of India.

Recognised as a Google for Education Partner.

ISO 9001:2015 certified organisation.

Headquartered at Chandivali IT Hub, Mumbai, India, with a hybrid, pan-India internship delivery model.

Website: www.codectechnologies.in | Email: support@codectechnologies.in

Codec Technologies offers skill-based internship programmes in emerging technology domains such as Artificial Intelligence, Machine Learning, Web Development, and Data Science, combining self-paced online modules with mentor-led project work and weekly performance evaluations

4.3 Problem Statement

With the explosive growth of user-generated text on social media, e-commerce platforms, and review sites, organisations receive enormous volumes of unstructured textual feedback every day. Manually reading and categorising this feedback to understand customer opinion is time-consuming, inconsistent, and does not scale with the volume of data being generated. There is, therefore, a need for an automated system that can accurately analyse a piece of text and determine the sentiment — positive, negative, or neutral — expressed within it, so that businesses can quickly gauge public opinion and respond accordingly.

4.4 Project Objectives

- To understand and apply core Natural Language Processing (NLP) concepts such as tokenisation, text cleaning, and normalisation.
- To preprocess raw textual data and convert it into a numerical form suitable for machine learning models.
- To build and train a classification model capable of predicting the sentiment of a given piece of text.
- To evaluate the model's performance using standard metrics such as accuracy, precision, recall, and F1-score.
- To gain hands-on experience with Python libraries used in AI/ML and NLP development (NumPy, Pandas, Scikit-learn, NLTK/SpaCy).

4.5 Scope of the Project

The scope of this project is limited to building and evaluating a text-based sentiment classification model using a labelled dataset (such as product or movie reviews) as part of a one-month learning internship. It covers data collection, text preprocessing, feature extraction, model training, and evaluation within a Jupyter/Colab notebook environment.

The project does not extend to deploying the model as a production-grade, real-time web service, nor does it cover multilingual sentiment analysis or advanced transformer-based architectures, which are noted as potential areas for future enhancement.

4.6 Technologies and Tools Used

- Programming Language: Python 3

- Development Environment: Jupyter Notebook / Google Colab
- Data Handling: NumPy, Pandas
- Data Visualisation: Matplotlib, Seaborn
- NLP Libraries: NLTK, SpaCy
- Machine Learning: Scikit-learn (Logistic Regression, TF-IDF Vectorizer, train-test split, evaluation metrics)
- Deep Learning (optional extension): TensorFlow / Keras Version Control & Documentation: Git, Markdown

4.7 System Architecture

The Sentiment Analysis system follows a standard NLP pipeline architecture, where raw text data flows through successive stages of cleaning, transformation, and modelling before producing a final sentiment label. The high-level architecture is summarised below:

- Data Input Layer → Raw text dataset (reviews / feedback / comments).
- Preprocessing Layer → Lowercasing, punctuation and stop-word removal, tokenisation, stemming/lemmatisation.
- Feature Extraction Layer → Conversion of cleaned text into numerical vectors using Bag-of-Words / TF-IDF.
- Model Layer → Classification algorithm (Logistic Regression) trained on labelled sentiment data.
- Evaluation Layer → Model assessment using accuracy, precision, recall, and F1-score on a held-out test set.
- Output Layer → Predicted sentiment label (Positive / Negative / Neutral) for new, unseen text.

Diagrammatically: Raw Text → Preprocessing → Feature Extraction (TF-IDF) → Model Training → Evaluation → Sentiment Prediction.

4.8 Methodology

The project was carried out in a sequence of well-defined steps, consistent with the standard Machine Learning / NLP workflow taught during Weeks 1–4 of the internship curriculum:

- Data Collection: Gathering a labelled text dataset (e.g., product/movie reviews) with associated sentiment labels.

- Data Cleaning: Handling missing values, removing duplicates, and normalising text.
- Exploratory Data Analysis (EDA): Visualising class distribution and word frequencies using Matplotlib/Seaborn.
- Text Preprocessing: Tokenisation, stop-word removal, and stemming/lemmatisation using NLTK/SpaCy.
- Feature Engineering: Transforming text into numerical features using TF-IDF vectorisation.
- Train-Test Split: Dividing the dataset into training and testing subsets.
- Model Building: Training a Logistic Regression classifier on the training data using Scikit-learn.
- Model Evaluation: Measuring performance using accuracy, precision, recall, F1-score, and a confusion matrix.
- Hyperparameter Tuning: Improving performance through cross-validation and parameter adjustment.
- Documentation & Presentation: Compiling the workflow, results, and learnings into the final capstone submission.

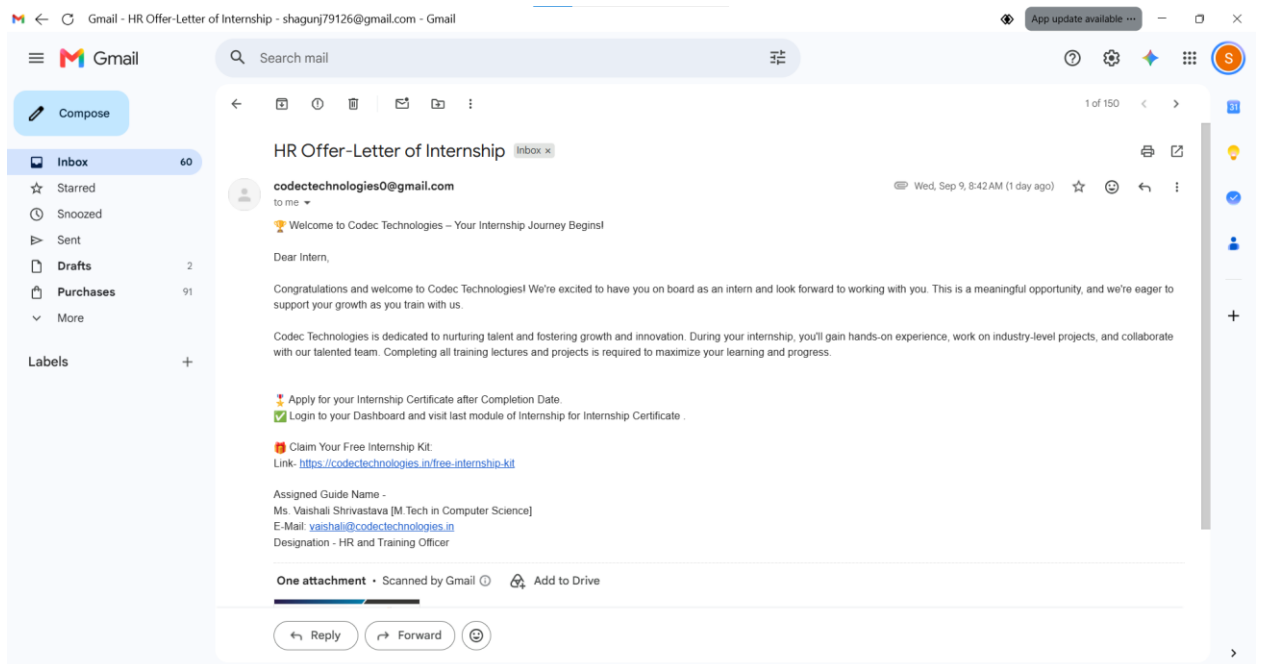
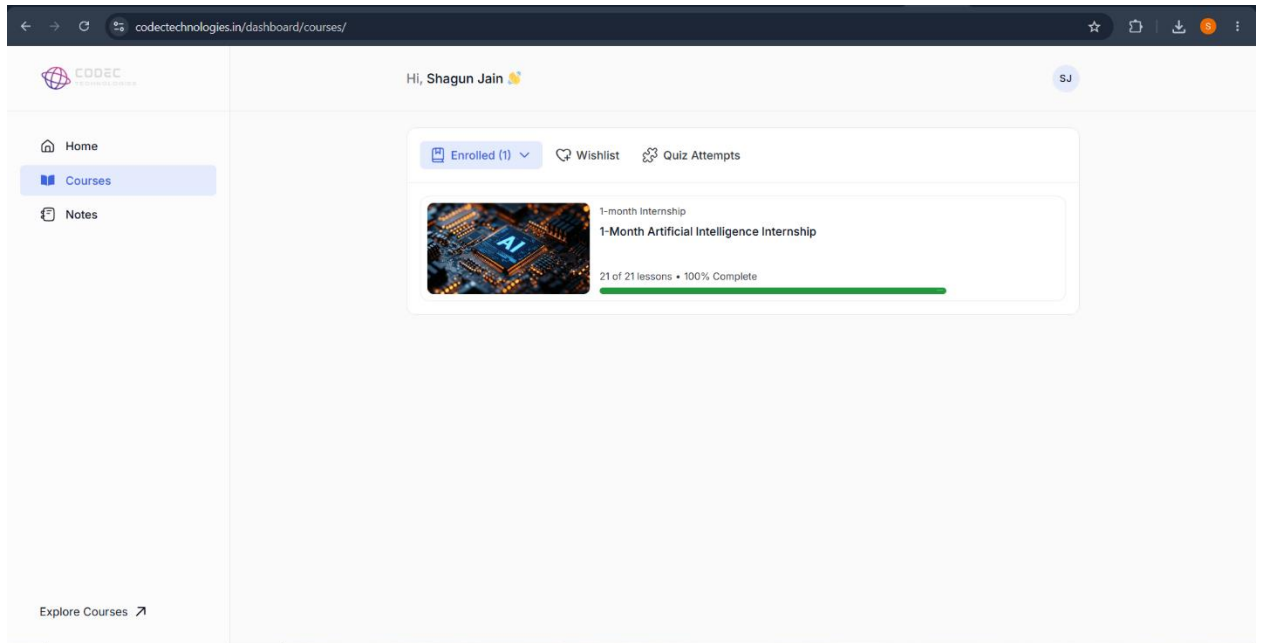
4.9 Expected Outcomes

- A trained sentiment analysis model capable of classifying text into positive, negative, or neutral categories with reasonable accuracy.
- A clear understanding of the end-to-end NLP and Machine Learning pipeline, from raw data to a working model.
- Practical proficiency in Python libraries for data handling, visualisation, NLP, and Machine Learning.
- Improved ability to evaluate and interpret model performance using standard classification metrics.
- A completed capstone project that consolidates all four weeks of internship learning, submitted for review by the assigned project head(s).
- Successful completion of the internship, reflected in the award of the Certificate of Internship.

4.10 Certificates of Completion and Communication Proof

On successful completion of all 21 course lessons and the capstone project, the internship dashboard reflected 100% course completion, and the Internship Completion Certificate (shown

in Section 3) was issued by Codec Technologies Pvt. Ltd., confirming the successful conduct of the internship from 22/07/2026 to 22/08/2026.



All assigned projects and training tasks were completed and submitted on the Codec Technologies platform and via email communication with the assigned project head(s), in accordance with the terms outlined in the internship offer letter.

5. BIBLIOGRAPHY/REFERENCES

- Codec Technologies Pvt. Ltd. – Official Website, <https://www.codectechnologies.in>
- Codec Technologies – 1-Month Artificial Intelligence Internship Course Curriculum (Internal Course Material).
- Scikit-learn Documentation, <https://scikit-learn.org/stable/documentation.html>
- NLTK (Natural Language Toolkit) Documentation, <https://www.nltk.org/>
- SpaCy Documentation, <https://spacy.io/>
- Pandas Documentation, <https://pandas.pydata.org/docs/>
- NumPy Documentation, <https://numpy.org/doc/>
- Matplotlib Documentation, <https://matplotlib.org/stable/contents.html>
- TensorFlow / Keras Documentation, <https://www.tensorflow.org/>
- National Internship Portal, Ministry of Education (AICTE), Government of India, <https://internship.aicte-india.org/>